

CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 21: Solids

Every formula and result of the chapter on one sheet

1	Cuboid, volume	lbh
2	Cuboid, total surface area	$2(lb + bh + hl)$
3	Cuboid, four walls	$2(l + b)h$
4	Cuboid, diagonal	$\sqrt{l^2 + b^2 + h^2}$
5	Cube, volume	a^3
6	Cube, total surface area	$6a^2$
7	Cube, diagonal	$a\sqrt{3}$
8	Open box	$lb + 2(l + b)h$
9	Uniform cross-section, volume	Area of cross-section $\times$ length
10	Uniform cross-section, lateral area	Perimeter of cross-section $\times$ length
11	Flow	Area of cross-section $\times$ speed $\times$ time
12	Capacity	1 cu m = 1000 litres; 1 litre = 1000 cu cm
13	Hollow box	Internal dimension = external $- 2 \times$ thickness
14	Number of small solids	Volume of large solid $\div$ volume of one small solid